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Machine Learning

Assignment - 2

SDG - Based Questions - CO2

rchitecture and Formulation:-

» Inputs x_1 = pH, x_2 = Turbidity (NTU),
x_3 = Dissolved oxygen (DO in mg/L).

→ Output: $y \in \{1$ (safe), 0 (unsafe)$\}$.

→ Net Input:- $z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = w^T x$
(where $x_0 = 1$ is the bias Input and w_0 is the bias weight)

→ Activation Function:- Unit step Function:

$$y = f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

2) weight Update Rule (Perceptron learning Rule):-
For a target label $y \in \{0, 1\}$ and learning rate
$\eta \in [0, 1]$:

$$\Delta w_i = \eta$$

w_i(new) = w_i(old) + Δw_i for $i \in \{0, 1, 2, 3\}$

→ If $\hat{y} = y$ (correct prediction): $\Delta w_i = 0$, weight remain unchanged.

→ If $y = 1$, $\hat{y} = 0$ (False Negative): weights associated with positive inputs are increased to raise z.

→ If $y = 0$, $\hat{y} = 1$ (False positive): weights are decreased to lower

2) Soil Fertility classification:

Derivation & Convergence:-

1) Derivation of perceptron rule:-

Let input vector

$x = [1$, Nitrogen (N), phosphorus (P), potassium (Pa) & weight

vector w.

→ Predict output $\hat{y} = \text{sgn}(w^T x)$, mapped to class $y \in \{+1$

(Fertile), -1 (Degraded) y

→ The error energy is given by the error (criter

function $J_p(w) = -\sum_{x \in \mu} w^T x y$,

where μ is the set of misclassified samples.

→ The gradient with respect to w is $\nabla J_p(w) = -\sum_{x}$

→ Applying stochastic gradient Descent (SGD) for a sin

misclassified sample :-

$$w^{(k+1)} = w^{(k)} - \eta \nabla J_p = w^{(k)} + \eta (y - \hat{y}) x$$

2) Convergence Conditions (Perceptron Convergence Theorem):-

1) Linear separability: The training data (soil samples) mu

be linearly separable in the 3D feature space (N,

i.e, there exists a hyper-plane w^* such that $y_i (w^* \cdot$

$\geq \gamma > 0$ for all samples.

2) Learning Rate condition: $0 < \eta \leq 1$ convergence occurs in

a finite no. of steps $k \leq \frac{R^2}{\gamma^2}$, where

$R = \max \| x_i \|$ and γ is the margin.

3) A single perceptron can learn only in linear decision boundary. It cannot correctly classify all data when the relationship between inputs and outputs is non-linear.

Ex:- Energy poverty may depend on house hold, income electricity availability, and energy prices, Different combinations of these factors can create complex patterns that cannot be separated by a straight line.

learn complex non-linear relationship
has no hidden layers to extract intermediate features.
→ May produce incorrect classification for complex datasets.

A multilayer perceptron (MLP) use hidden layers and non-linear activation function such as ReLU. It can learn more complex relationships and is therefore more suitable for energy-poverty classification when the data is non-linearly separable.

A perceptron can predict malnutrition risk using BMI and dietary intake as inputs. It's output is 1 for at risk and 0 for not at risk.

The weight update formula is:-

$$w_i(new) = w_i(old) + \eta(t-y)x_i$$

Assume initial weight $w_1 = w_2 = 0$ bias $b=0$ learning rate $\eta = 0.5$ and a prediction threshold of 0.5.

Iteration	Inputs	Target	Prediction	updated weights.
1)	(1, 1)	1	0	$w_1=0.5, w_2=0.5$
2)	(0, 1)	0	1	$w_1=0.5, w_2=0$

Explanation:-

→ Iteration 1: The prediction is incorrect, so both weight increase by 0.5.

→ Iteration 2:- The prediction in incorrect, so the second weight decreases by 0.5.

The process repeats until the model learns the training patterns. This example illustrates the algorithm only; actual malnutrition assessment requires appropriate nutritional.

5) A perceptron classifies air quality as hazardous or safe using pollutant measurements such as PM2.5, nitrogen dioxide, and ozone.

weight sum is:- $z = \sum_{i=1}^{n} w_i x_i + b$

The activation function converts the result into a binary output. However, a single perceptron can represent only a linear decision boundary.

Limitations: 1) It cannot learn complex non-linear relationships b/w pollutants.

2) It has limited ability to represent interactions between multiple pollutants.

3) It may misclassify samples when hazardous conditions depend on combinations of pollouteants and wheather.

4) It's performance depends on the quality of the training data.

Ex: High traffic emissions combined with low wind speed may cause complex pollution patterns that a single perceptron cannot classify accurately.

A multilayer perceptron can overcome these limitations using hidden layers and non-linear activation function.